

Bluestock Mutual Fund Analytics Capstone

Final Project Report

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Tools: Python, pandas, SQLite, Power BI Service, Jupyter Notebook

Version: v1.0

1. Executive Summary

This project builds an end-to-end mutual fund analytics system for Bluestock. It transforms raw fund, NAV, SIP, AUM, investor transaction, benchmark, and portfolio holding data into cleaned datasets, a SQLite warehouse, analytical notebooks, risk metrics, recommender logic, and a four-page interactive dashboard.

Metric	Value
Funds covered	40
NAV records	64,320
Investor transactions	32,778
Latest AUM	■62.74 lakh crore
Latest SIP inflow	■31,002 crore
Latest folios	26.12 crore
Schemes tracked	1,522

2. Data Sources

- Fund master: AMFI code, fund house, scheme name, category, plan, risk grade, benchmark, and expense ratio.
- NAV history: daily NAV values used for returns, CAGR, VaR, CVaR, Sharpe, Sortino, drawdown, and benchmark comparison.
- AUM and SIP data: industry-level trend tables for AUM growth, SIP inflows, folios, and category flows.
- Investor transactions: investor-level transaction date, type, amount, state, city tier, age group, gender, and KYC status.
- Portfolio holdings and benchmarks: sector allocation, HHI concentration, Nifty 50 and Nifty 100 comparison series.

3. ETL Design

The ETL pipeline reads raw CSVs from data/raw, standardises dates and numeric values, removes duplicates, validates business rules, and writes cleaned outputs to data/processed. NAV history is sorted by AMFI code and date, invalid NAV values are removed, and missing calendar dates are forward-filled to handle weekends and holidays.

- NAV validation: date parsing, duplicate removal, NAV > 0 checks, and forward-fill after full date reindexing.
- Transaction validation: transaction type standardisation, amount > 0, KYC status checks, and date parsing.
- Performance validation: numeric return checks, expense ratio range checks, and anomaly flags.
- SQLite load: SQLAlchemy writes cleaned data to star-schema tables and verifies row counts.

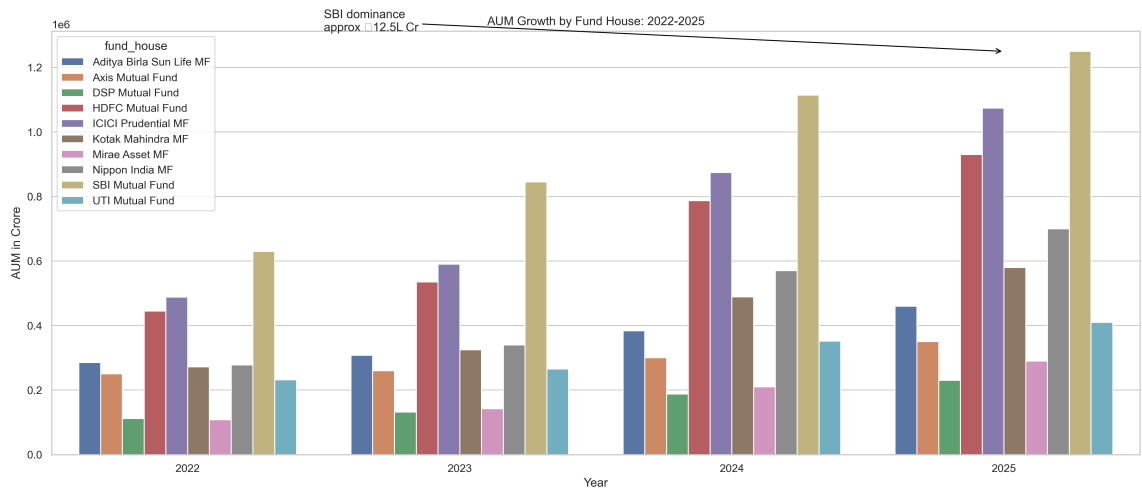
4. SQLite Star Schema

The warehouse is designed around dimension and fact tables. `dim_fund` stores the fund master attributes and `dim_date` stores reusable calendar fields. Fact tables store NAV, transactions, performance, AUM, SIP inflow, category inflow, folio count, portfolio holdings, and benchmark values.

- Primary keys: `dim_fund.amfi_code` and `dim_date.date`.
- Main foreign keys: `fact_nav.amfi_code`, `fact_transactions.amfi_code`, `fact_performance.amfi_code`.
- SQL files: `sql/schema.sql` and `sql/queries.sql` document the database structure and analysis queries.

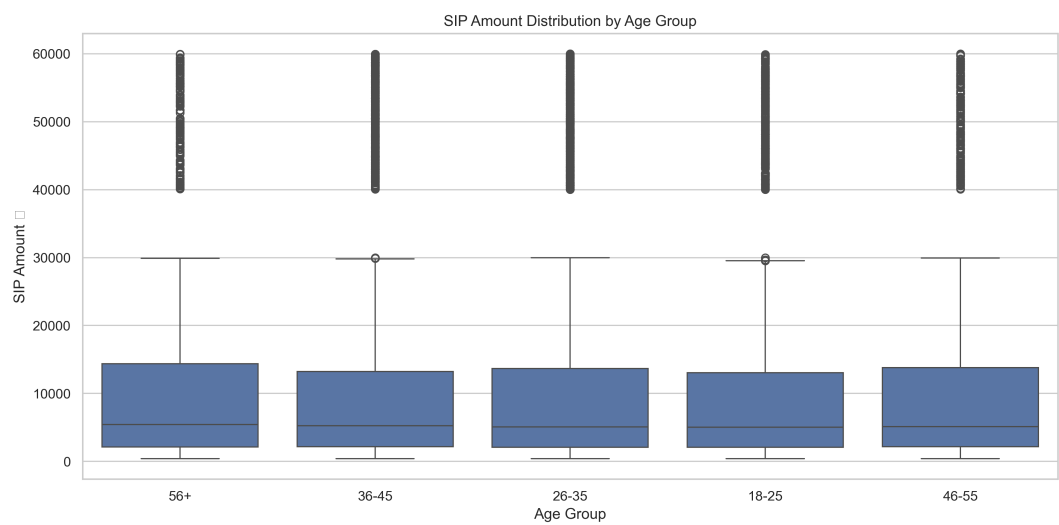
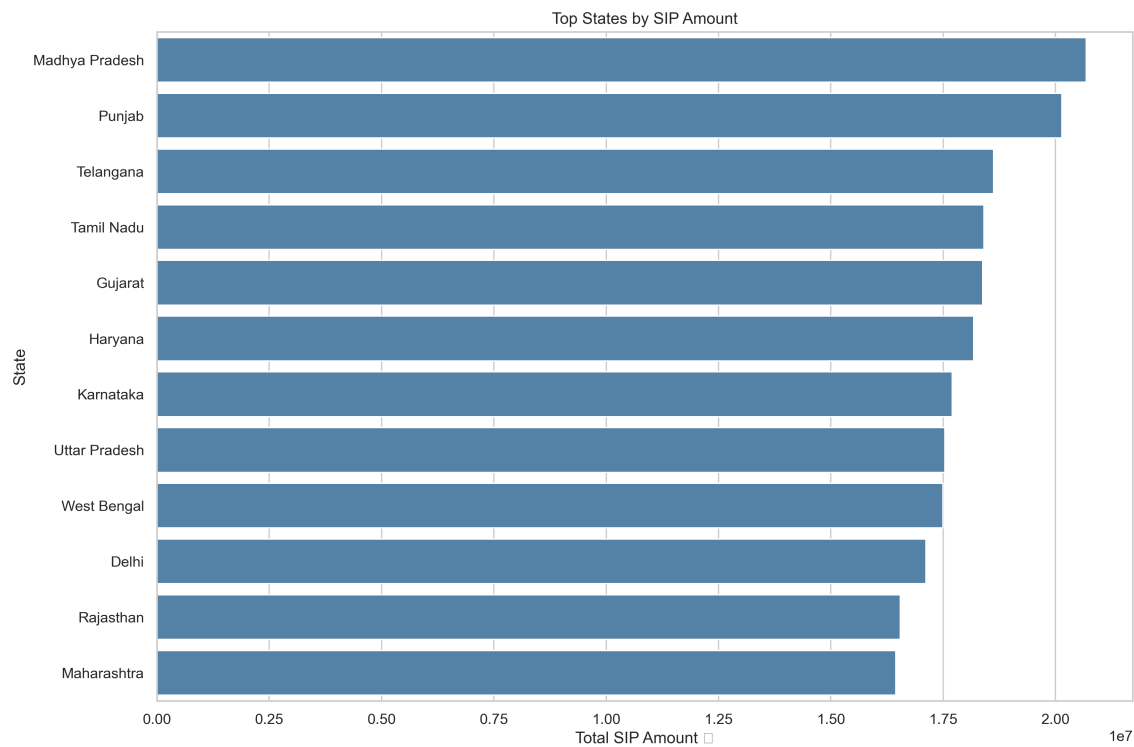
5. EDA Findings: Industry and SIP Trends

EDA shows a strong upward trend in SIP participation and industry-level AUM. The dashboard highlights fund house dominance and month-level variation in category inflows.



6. EDA Findings: Investor and Geography Patterns

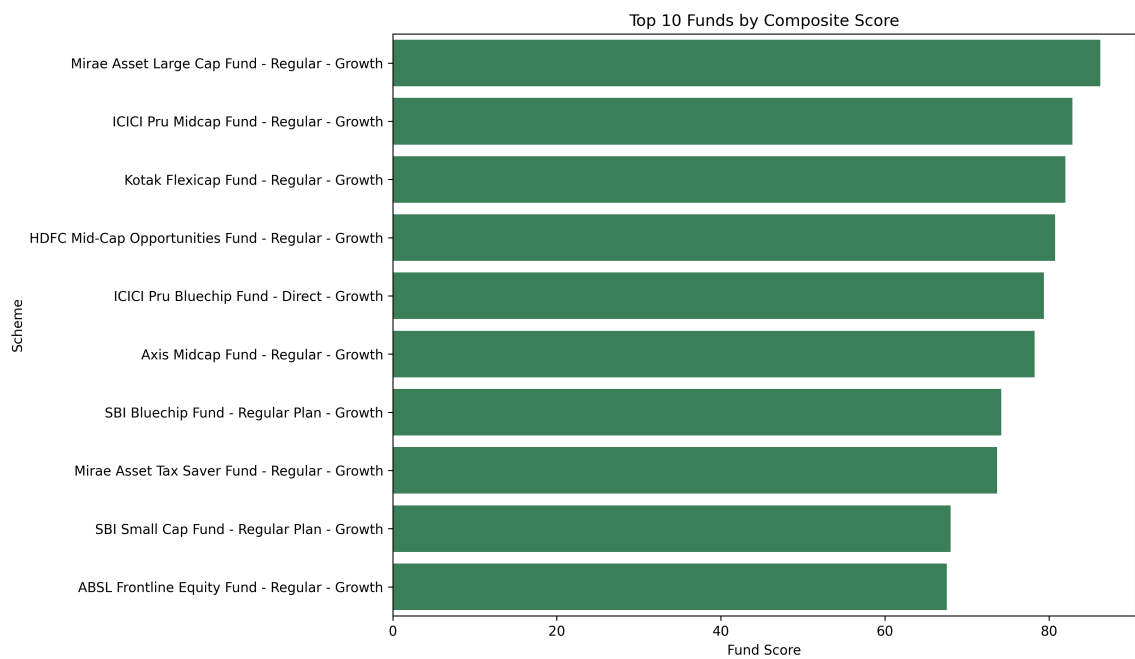
Investor analytics reveal meaningful differences across state, age group, city tier, and transaction type. SIP amount patterns vary by investor age group, while state-wise contribution is concentrated in major markets.



7. Performance Analysis

Performance analytics cover CAGR, Sharpe ratio, Sortino ratio, Alpha, Beta, maximum drawdown, tracking error, and a composite scorecard.

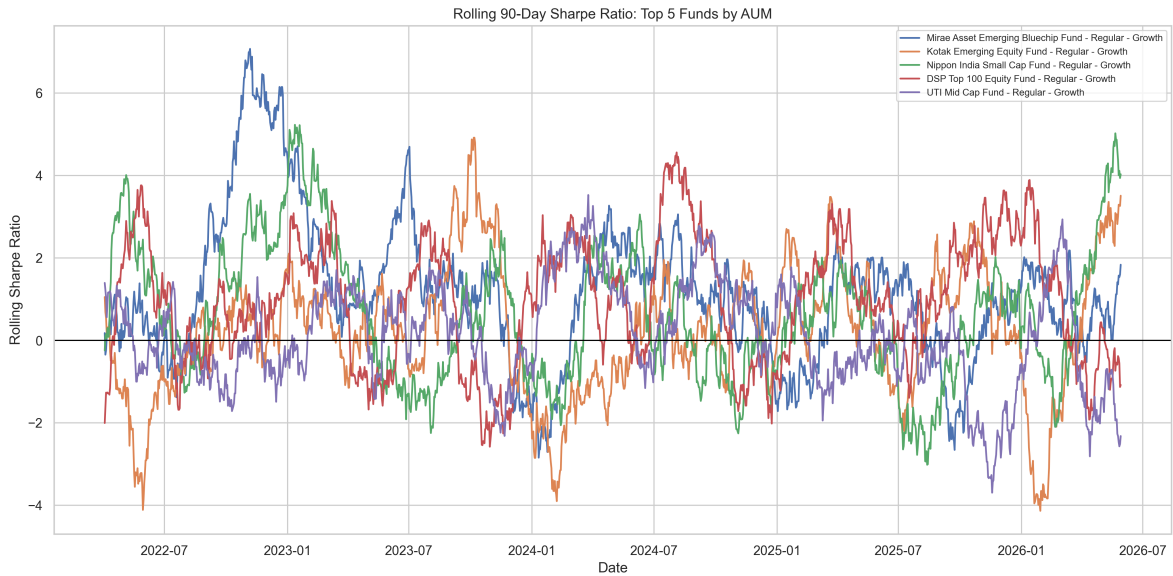
scheme_name	fund_house	cagr_3yr	sharpe_ratio	alpha_annual	fund_score
Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	0.340	1.068	0.270	86.250
ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF	0.318	0.883	0.293	82.875
Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF	0.296	0.966	0.273	82.000
HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund	0.324	0.808	0.272	80.750
ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF	0.325	0.715	0.212	79.375
Axis Midcap Fund - Regular - Growth	Axis Mutual Fund	0.351	0.731	0.261	78.250



8. Advanced Risk Analytics

The advanced analytics notebook computes Historical VaR and CVaR at 95%, rolling 90-day Sharpe, cohort behavior, SIP continuity risk, recommender outputs, and sector HHI concentration.

scheme_name	fund_house	var_95	cvar_95	daily_volatility
ABSL Small Cap Fund - Regular - Growth	Aditya Birla Sun Life MF	-0.024	-0.030	0.014
Axis Small Cap Fund - Regular - Growth	Axis Mutual Fund	-0.023	-0.030	0.013
SBI Small Cap Fund - Direct Plan - Growth	SBI Mutual Fund	-0.023	-0.030	0.013
Nippon India Small Cap Fund - Regular - Growth	Nippon India MF	-0.023	-0.030	0.013
DSP Small Cap Fund - Regular - Growth	DSP Mutual Fund	-0.022	-0.029	0.013
SBI Small Cap Fund - Regular Plan - Growth	SBI Mutual Fund	-0.022	-0.028	0.013

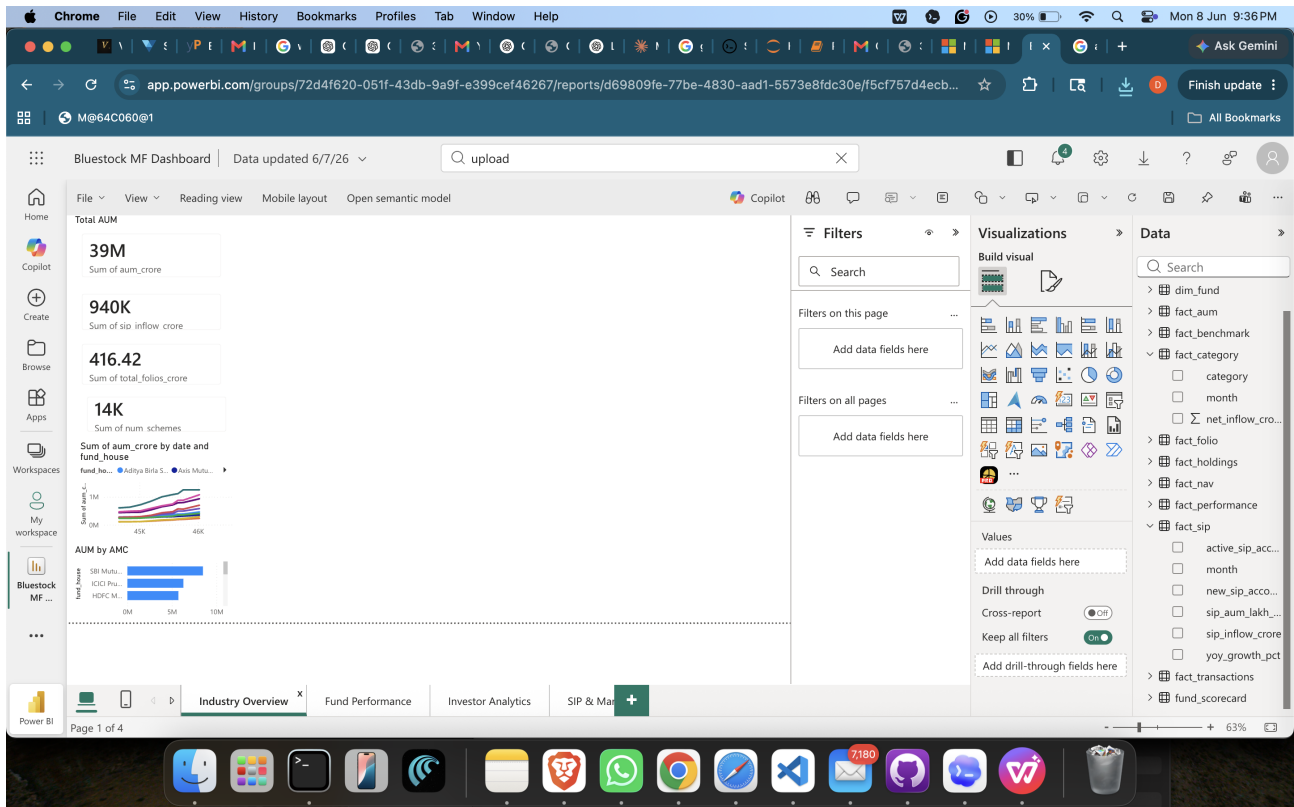


9. Dashboard Overview

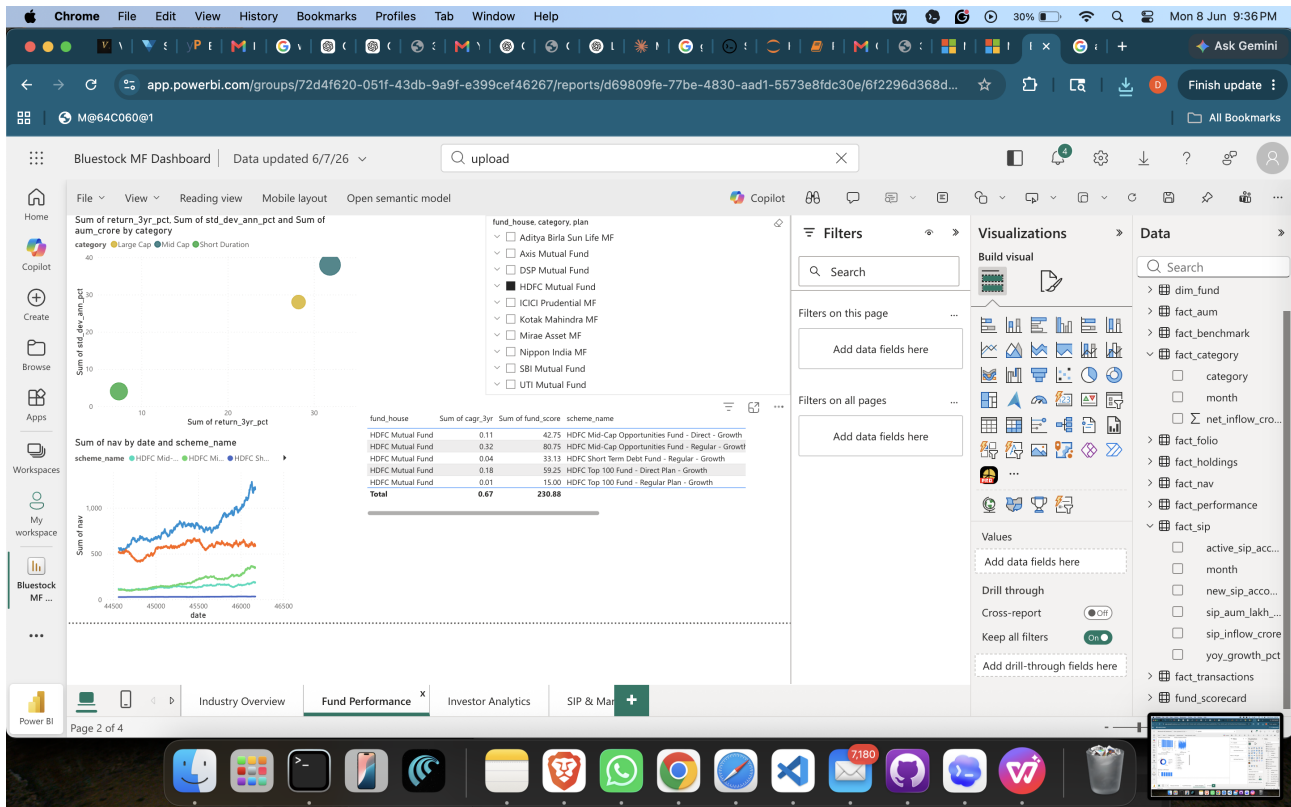
The dashboard was built in Power BI Service on Mac and contains four report pages: Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends.

- Industry Overview: Total AUM, SIP inflow, folios, schemes, AUM trend, and AUM by AMC.
- Fund Performance: return-risk scatter, fund scorecard, NAV trend, and slicers.
- Investor Analytics: state-level transactions, transaction type split, age group SIP amount, and monthly volume.
- SIP & Market Trends: SIP inflow trend, Nifty 50 trend, and category inflow heatmap.

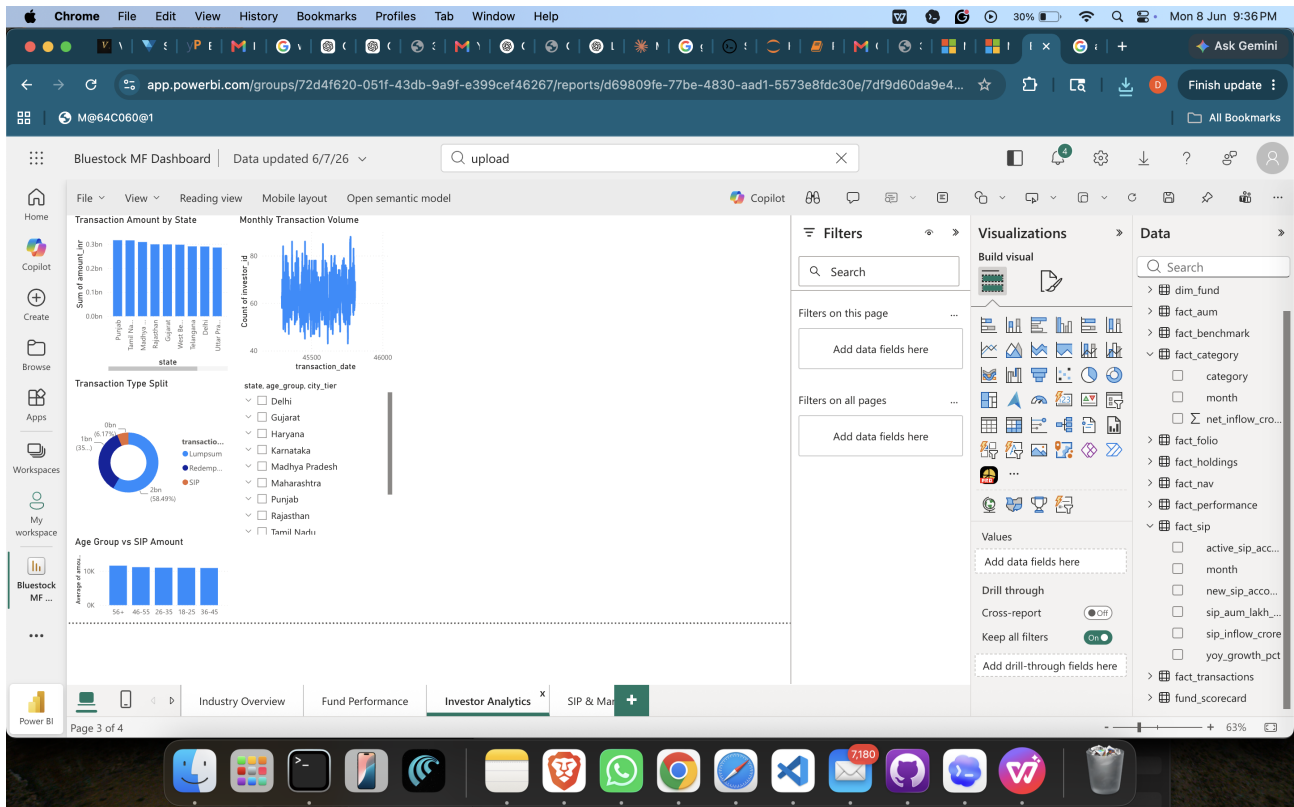
10. Dashboard Screenshot: Industry Overview



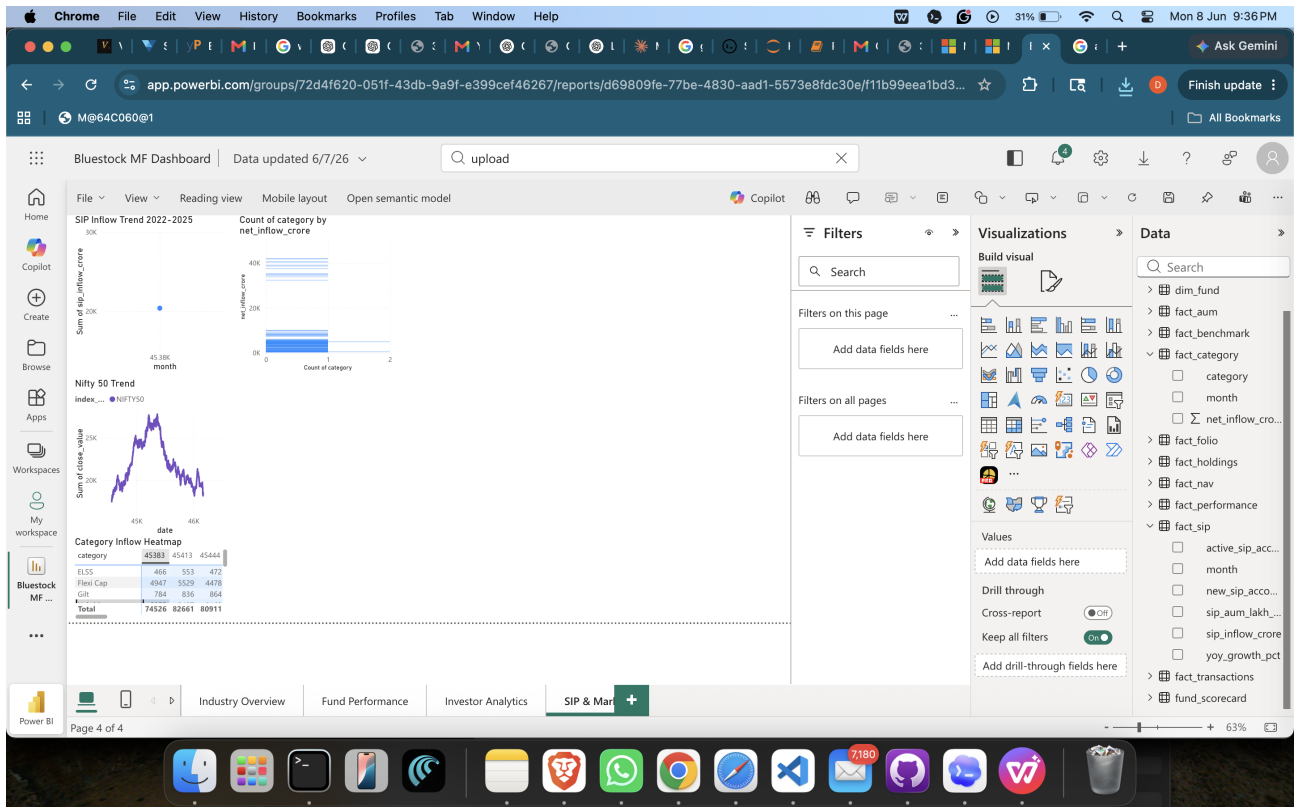
11. Dashboard Screenshot: Fund Performance



12. Dashboard Screenshot: Investor Analytics



13. Dashboard Screenshot: SIP & Market Trends



14. Limitations

- Power BI Desktop is not available natively on Mac, so the dashboard was built in Power BI Service.
- The browser workflow can export PDF and screenshots, but PBIX creation requires Windows Power BI Desktop.
- Some analysis outputs depend on the quality and completeness of the provided source datasets.
- SQLite database files are excluded from GitHub to keep the repository clean; schema.sql documents the database design.

15. Recommendations

- Use scorecard ranking as a first-pass screening tool, then review fund mandate and portfolio concentration.
- Monitor SIP continuity flags to identify investors who may need engagement before drop-off.
- Use VaR and CVaR together because CVaR captures average losses during the worst return days.
- Track expense ratios alongside returns because low-cost funds with strong risk-adjusted returns improve long-term outcomes.
- Extend the project with scheduled NAV refresh, Streamlit dashboard, Monte Carlo simulation, and portfolio optimization.

16. Self-Review Checklist

Objective	Status
ETL pipeline	Complete
SQLite schema and SQL queries	Complete
EDA notebook and charts	Complete
Performance analytics and CSV outputs	Complete
Power BI dashboard pages	Complete in Power BI Service
Advanced analytics and recommender	Complete
Final report and presentation	Complete

The project is ready for final GitHub submission with a v1.0 tag.